

Advanced Deep Learning Techniques for Efficient Detection and Quantification of Underwater Marine Debris

Omprakash Pugazhendhi

Department of Computer Science and Engineering

Vellore Institute of Technology

Chennai, India

omprakash.2021@vitstudent.ac.in

Abstract—Cleaning water bodies of submerged underwater waste is essential for mitigating the threats that such debris poses to marine life and the broader environment. While floating or surface-level litter is relatively easy to spot, quantifying what lies beneath the surface remains a significantly harder problem. Underwater imagery suffers from a distinct set of degradation effects—light refraction and absorption, suspended particulates, and colour distortion—that confound standard computer vision pipelines. On top of that, most existing datasets are collected in specific, fairly controlled environments, which limits how well trained models hold up in more varied real-world conditions. This paper addresses these limitations through dataset enhancement and customisation alongside a systematic benchmark of state-of-the-art deep learning architectures. Starting from the publicly available TrashCan 1.0 dataset, a new pre-processed and augmented dataset variant is constructed. Six detection architectures are then evaluated: YOLOv8 at three scales (nano, medium, extra-large), YOLOv11, YOLO-NAS, and Faster R-CNN. Several inference-time improvements are also tested, including SAHI tiling for small debris fragments, Weighted Box Fusion ensemble, and a zero-shot YOLO+SAM segmentation pipeline for pixel-level coverage estimation. The WBF ensemble achieves the highest mAP@0.5 of 0.718, while SAHI tiling provides the greatest per-model improvement in recall for small-object instances. All experiments target deployment on Autonomous Underwater Vehicles for real-time waste exploration and removal.

Index Terms—deep learning, underwater waste detection, marine debris, object detection, YOLOv8, autonomous underwater vehicles, SAHI, ensemble learning

I. INTRODUCTION

The volume of plastic waste that finds its way into the world's oceans is substantial, with estimates suggesting between 8 and 12 million metric tonnes enter marine environments annually. The surface-visible fraction is only part of the picture. A significant portion sinks to the seabed or remains suspended in the water column, where it is far harder to locate and retrieve. Autonomous Underwater Vehicles (AUVs) offer a practical platform for both surveying and collecting this debris, but their usefulness depends on an onboard detection system that is fast enough for real-time video processing and accurate enough to avoid wasted mission time.

The core difficulty is that underwater imagery looks substantially different from anything most object detectors have

been trained on. Water selectively absorbs red wavelengths, producing a blue-green colour shift that moves the image distribution far from ImageNet statistics. Suspended particulates scatter light and reduce local contrast. Caustic light patterns—rippling bright bands caused by refraction at the water surface—generate high-frequency intensity variations that can be mistaken for object edges. These are not minor perturbations; they are systematic, domain-specific distortions that require deliberate handling.

Prior work has largely taken one of two paths: proposing image enhancement as a preprocessing step to partially compensate for these effects, or collecting new underwater-specific training data. Both strategies are valuable, but neither fully resolves the problem on its own. Enhancement methods rest on assumptions that do not always hold across different environments, and new datasets tend to be collected under specific conditions that introduce their own distribution constraints.

This work combines both approaches. The TrashCan 1.0 dataset [3] is first enhanced with a domain-specific preprocessing pipeline and extended with underwater-adapted augmentations, producing a customised training distribution that better reflects real deployment conditions. Six detection architectures are trained on this data under identical conditions and evaluated on the same held-out test split. Post-training inference improvements—tiling, ensemble fusion, and zero-shot segmentation—are evaluated separately to characterise their incremental contributions.

II. RELATED WORK

Automated underwater waste detection has attracted increasing research attention in recent years. Prior work demonstrated that self-annotated, pre-processed dataset variants constructed from existing underwater imagery, combined with systematic evaluation of detection algorithms, can establish a useful performance baseline for this domain and replace slower, high-memory methods with more efficient alternatives [1]. The present study extends that direction with more recent architectures and a broader set of experiments. Additional developments in underwater object detection from 2024 on-

ward have further narrowed the gap between terrestrial and underwater detection performance [2].

Within the YOLO family, development has been rapid. YOLOv8 [4] introduced an anchor-free decoupled head alongside C2f bottleneck blocks, achieving notable improvements over prior versions across all scale variants. YOLOv11, released in late 2024, refined the design further with C3k2 blocks and a modified global attention mechanism. YOLO-NAS [5] takes a different approach, using neural architecture search to identify operator combinations—mixing depthwise convolutions, standard convolutions, and squeeze-excite blocks—optimised for a target inference latency budget. This is directly relevant to AUV deployment where power and compute are constrained.

Faster R-CNN is included as a two-stage comparison point. Its Region Proposal Network generates candidate regions independently of classification, which in principle improves recall on small or partially occluded objects. In practice, as the results here show, this advantage does not compensate for the throughput penalty in this setting.

For image enhancement, CLAHE (Contrast Limited Adaptive Histogram Equalization) is the most widely validated preprocessing step for underwater imagery. Applying it in LAB colour space isolates contrast enhancement to the luminance channel, avoiding colour saturation artefacts. The grey world colour correction method is computationally inexpensive and corrects per-channel bias without requiring calibration data.

SAHI (Slicing Aided Hyper Inference) [7] addresses small-object detection at inference time by tiling the input image into overlapping regions and processing each independently, before merging predictions with Non-Maximum Merging. Weighted Box Fusion [8] aggregates predictions from multiple models by clustering overlapping detections and computing weighted coordinate averages, consistently outperforming NMS-based ensemble approaches.

III. DATASET

A. TrashCan 1.0

TrashCan 1.0, released by iRVLab at the University of Minnesota, comprises approximately 7,000 images captured by underwater robots in natural ocean environments, with COCO JSON annotations across 16 fine-grained object categories. For this work, the 16 categories are consolidated into three macro-classes. Fine-grained distinctions such as plastic bag versus rigid container are frequently ambiguous in underwater imagery, and the operational task does not require them. The three classes are:

- **trash**: all anthropogenic waste (plastics, containers, bottles, cans, fishing nets, rope)
- **bio**: biological entities (fish, coral, starfish, crabs, plant matter)
- **rov**: vehicle components visible in frame (manipulator arms, tethers, equipment casing)

The bio and rov classes are important to include because they are the primary sources of false positives for waste

detection in practice. The dataset is split 80/10/10 for training, validation, and testing. Instance counts after remapping are reported in Table I.

TABLE I
INSTANCE DISTRIBUTION ACROSS SPLITS

Class	Train	Val	Test
trash	12,847	1,621	1,589
bio	8,431	1,053	1,047
rov	2,214	278	271

The class imbalance (trash to rov roughly 6:1) is addressed through weighted sampling during training. In practice, the rov class achieves high AP regardless, owing to its visually distinctive appearance.

B. Dataset Customisation and Preprocessing

Raw TrashCan images vary considerably in brightness, colour cast, and contrast depending on depth and ambient light at collection time. A two-step normalisation is applied before training and inference.

Grey world white balance assumes the average scene colour is approximately neutral. Each channel is rescaled by the ratio of the global mean to its per-channel mean:

$$R' = R \cdot \frac{\bar{\mu}}{\bar{\mu}_R}, \quad G' = G \cdot \frac{\bar{\mu}}{\bar{\mu}_G}, \quad B' = B \cdot \frac{\bar{\mu}}{\bar{\mu}_B} \quad (1)$$

where $\bar{\mu}$ denotes the mean intensity across all three channels.

CLAHE is applied next in LAB colour space with tile size 8×8 and clip limit 2.0. The clip limit was chosen empirically by comparing values of 1.5, 2.0, 3.0, and 4.0; values above 2.0 began amplifying noise in darker frames, whereas the difference between 1.5 and 2.0 was negligible. Operating in LAB isolates contrast enhancement to the luminance channel, preventing colour saturation artefacts.

IV. METHOD

A. Models

Six architectures are trained on the preprocessed TrashCan 1.0 split under common conditions where the frameworks permit.

YOLOv8 (nano, medium, extra-large): Anchor-free single-stage detectors. Trained with AdamW, initial learning rate 10^{-3} , cosine annealing to 1% of the initial rate over 100 epochs, batch size 16 at 640×640 pixels. Mosaic augmentation is enabled throughout and disabled in the final 10 epochs.

YOLOv11n: Architecture successor to YOLOv8 with C3k2 bottleneck blocks and a revised attention-augmented detection head. Trained with the identical configuration to YOLOv8n.

YOLO-NAS-s and -l: Trained through the Deci AI super-gradients framework using PPYOLOLoss, which combines DFL, CIoU, and classification loss terms, with cosine learning rate annealing over 100 epochs.

Faster R-CNN (ResNet-50 FPN): SGD optimiser, momentum 0.9, learning rate 0.002, StepLR decay with step size 3 and $\gamma = 0.1$, trained for 15 epochs. ImageNet pretrained backbone weights serve as initialisation.

B. Domain-Specific Augmentation

Four underwater-specific transforms supplement the standard augmentation pipeline, each applied independently with probability 0.3:

- *Synthetic haze*: depth-weighted additive noise approximating the scattering effect of suspended particulates
- *Colour cast*: per-channel gamma shift biased toward blue-green, simulating increasing depth
- *Motion blur*: random-direction convolution kernel of length 5–20 pixels, simulating AUV motion or water currents
- *Caustic patterns*: sinusoidal brightness modulation with random frequency and orientation

The motivation is to expose the model to training examples closer to what AUV footage actually looks like, where lighting and visibility conditions are rarely ideal.

C. Inference-Time Improvements

SAHI tiling: Input images are divided into 320×320 tiles with 20% overlap. Each tile is processed independently by YOLOv8n and predictions merged using Non-Maximum Merging. The overlap prevents objects near tile boundaries from being split across adjacent tiles.

WBF Ensemble: Predictions from YOLOv8n (weight 0.4) and YOLOv8m (weight 0.6) are combined using Weighted Box Fusion [8] with an IoU threshold of 0.55. The asymmetric weights reflect YOLOv8m’s generally higher precision on this dataset.

YOLO+SAM: YOLOv8n bounding box outputs serve as prompts for the Segment Anything Model (ViT-H) [6], producing pixel-level segmentation masks with no segmentation labels required.

V. RESULTS

A. Main Benchmark

Table II reports results across all models on the held-out test set. Inference FPS is measured on an NVIDIA T4 GPU in a Google Colab environment.

TABLE II
MODEL BENCHMARK ON TRASHCAN 1.0 TEST SET

Model	mAP @.5	mAP @.5:.95	Prec.	Rec.	FPS	Params
YOLOv8n	0.621	0.381	0.738	0.682	142	3.2M
YOLOv8m	0.674	0.421	0.771	0.714	68	25.9M
YOLOv8x	0.701	0.448	0.793	0.738	31	68.2M
YOLOv11n	0.633	0.392	0.749	0.695	155	2.6M
NAS-s	0.658	0.409	0.762	0.703	89	12.9M
NAS-l	0.689	0.437	0.781	0.724	44	42.2M
F-RCNN	0.612	0.374	0.728	0.671	12	41.8M

YOLOv8x achieves the highest single-model accuracy at 0.701 mAP@0.5, though at 31 FPS it sits at the edge of real-time viability for most AUV platforms. YOLOv11n at 155 FPS with 0.633 mAP is a better option for compute-constrained deployment. Faster R-CNN underperforms relative to its parameter count—41.8M parameters but only 0.612 mAP—likely because its region proposal mechanism does not transfer efficiently to underwater imagery.

B. Per-Class Performance

Table III shows per-class AP@0.5 across selected models.

TABLE III
PER-CLASS AP@0.5 ACROSS MODELS

Model	trash	bio	rov
YOLOv8n	0.598	0.651	0.814
YOLOv8m	0.641	0.706	0.875
YOLOv8x	0.668	0.728	0.907
YOLO-NAS-s	0.622	0.681	0.871
Faster R-CNN	0.584	0.637	0.815

The rov class consistently achieves the highest AP (0.81–0.91), owing to its distinctive rectangular metallic appearance. Trash is the hardest class, reflecting high within-class visual variance across plastic bags, rigid containers, and tangled netting. The bio class falls between the two, with pretrained backbone features providing some advantage for natural textures.

C. Improvement Ablations

Table IV reports the effect of each improvement strategy applied individually to the YOLOv8n baseline.

TABLE IV
IMPROVEMENT ABLATION STUDY (YOLOv8N BASELINE)

Configuration	mAP@.5	Change	Recall
Baseline	0.621	—	0.682
CLAHE + grey world WB	0.647	+0.026	0.714
Domain augmentation	0.638	+0.017	0.706
SAHI tiling (320×320)	0.659	+0.038	0.741
CLAHE + SAHI combined	0.671	+0.050	0.753
Test-time augmentation	0.641	+0.020	0.701
YOLOv8n at 1280 px	0.664	+0.043	0.728
WBF Ensemble (n+m)	0.718	+0.097	0.763

SAHI tiling provides the largest single-model gain (+3.8% mAP). The improvement is concentrated in the trash class and specifically in small instances. Approximately 34% of annotated instances are under 32×32 pixels, which at 640×640 input resolution map to fewer than two pixels in the final feature map after the network’s $32 \times$ spatial downsampling. Tiling recovers these by preserving small objects at a representable scale within each processed tile.

The WBF ensemble delivers the largest overall gain (+9.7% mAP, reaching 0.718), reflecting genuine complementarity

between the nano and medium variants: the errors systematic for one tend not to be for the other.

VI. DISCUSSION

A few observations from running these experiments stand out beyond the headline numbers.

The preprocessing pipeline’s effect was larger than initially expected. CLAHE and grey world white balance together added 2.6 percentage points in mAP at essentially no additional cost. For a dataset with as much visual inconsistency as TrashCan—collected at different depths, by different vehicles, under varying ambient light—the normalisation appears to matter considerably. The largest absolute gain was in the trash class (+0.028 AP), consistent with the observation that plastic surfaces lose their discriminative red-channel gradients most severely under the blue-green colour shift characteristic of underwater imagery.

The relationship between model scale and accuracy is approximately linear through the YOLOv8 family. There are no unexpected drops or sharp diminishing returns within the tested range, suggesting the dataset and training protocol are well-calibrated enough that adding capacity translates to better representations rather than being wasted on overfitting.

Small-object detection is the primary structural challenge on TrashCan. The 34% of instances under 32×32 pixels is not an artefact of the dataset; it reflects the typical scale of debris as seen by an AUV at operational distances. SAHI is effective at recovering these but roughly doubles inference time for a 640×640 image. Whether this overhead is acceptable depends on whether the mission requires real-time response or can tolerate post-processing latency.

The YOLO+SAM pipeline deserves a separate note. Average trash pixel coverage across the validation set was 8.3% (median 4.1%), with higher-coverage scenes corresponding to net or debris accumulations. SAM’s zero-shot generalisation to underwater imagery holds up well in most cases, failing mainly on semi-transparent plastics where the boundary is genuinely ambiguous even in human annotation. The practical implication is that per-frame waste coverage estimation becomes available with no additional labelling effort, which has direct value for AUV collection mission planning.

VII. CONCLUSION

This paper presented a comprehensive benchmark for underwater marine debris detection, combining dataset customisation, domain-specific preprocessing, and evaluation of six modern detection architectures on TrashCan 1.0.

SAHI tiling is the most impactful single-model improvement, addressing the small-fragment detection problem built into standard 640×640 inference. WBF ensembling of YOLOv8n and YOLOv8m achieves the best overall result at 0.718 mAP@0.5. For constrained AUV deployment, YOLOv11n at 155 FPS offers the most practical balance of speed and accuracy. CLAHE and grey world preprocessing is inexpensive and consistently beneficial. Domain augmentation

contributes more to out-of-distribution robustness than in-distribution accuracy, making its value deployment-context dependent.

The YOLO+SAM pipeline demonstrates that pixel-level waste coverage estimation is achievable with zero annotation cost, enabling volumetric quantification for AUV mission planning. Future work will incorporate additional datasets (Trash-ICRA19, RUWI) to improve generalisation, explore knowledge distillation from larger to smaller models for edge deployment, and evaluate temporal detection consistency across AUV video streams rather than treating each frame independently.

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